AVIJIT BASAK

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CAREER OBJECTIVE

Final-year Industrial & Production Engineering student at BUET (CGPA 3.44) with a balanced foundation in engineering systems, operations, and management principles. Possesses practical exposure to manufacturing environments, process analysis, and structured problem-solving through industrial training. Committed to leveraging analytical skills and cross-functional knowledge to support organizational processes and drive responsible decision-making in a dynamic entry-level role.

EDUCATION

B.Sc. in Industrial & Production Engineering (IPE) *(Ongoing)*
Bangladesh University of Engineering and Technology (BUET)
Level 4, Term 2 | CGPA: **3.44 / 4.00**

HSC (2020) – Govt. Azizul Haque College, Bogura | GPA: **5.00 / 5.00**
SSC (2018) – BIAM Model School & College, Bogura | GPA: **5.00 / 5.00**

INDUSTRIAL EXPERIENCE

Certified Industrial Attachment | PRAN-RFL Group *(4 Weeks)*

- Rotational exposure across **Plastics, Dairy, Beverage (CSD), Confectionery, AMCL**
- Observed **Tetra Pak aseptic packaging, CIP systems, and injection molding**
- Conducted **time study** on secondary packaging → identified ~**20% excess labor utilization**
- Identified inefficiencies in **Pulse Candy mixing** and **manual Ghee Butter cooking**; proposed **automation-based improvements**
- Gained exposure to **end-to-end supply chain flow**

ENGINEERING PROJECTS

Automatic Coconut Scraper – Product Development

- 3D CAD & motion study (SolidWorks); FEA safety analysis (ANSYS)
- Prototype fabrication (motor setup, sheet metal, welding)
- Mitigated vibration through structural damping

Other Design Work

- **Taj Mahal 3D Model** – Advanced SolidWorks surface & assembly modeling
- **Architectural Layout** – 2D residential layout using AutoCAD

RESEARCH EXPERIENCE

Undergraduate Thesis – Air Quality Prediction using Machine Learning

- Satellite data processing using **Google Earth Engine (ERA5, MODIS, TROPOMI)**
- ML modeling with **XGBoost, LightGBM, and CatBoost** for PM2.5 estimation
- Focus on **Bangladesh air quality mapping** for monitoring and planning

TECHNICAL SKILLS

Design & Simulation: SolidWorks, AutoCAD, CATIA, ANSYS, Arena (basic), GEE
Programming & Data: Python, C, JavaScript, MATLAB (basic), Pandas, NumPy, Matplotlib
Tools: MS Word, Excel, PowerPoint, Project
Languages: Bengali (Native), English (Professional)

ACTIVITIES, TRAINING & ACHIEVEMENTS

- **Hult Prize 2025 (Global)**
- **Climate Justice Idea Competition (OPR S3)** – 2nd Round
- **Over the Wall (Merico)** – Semi-finalist
- **National STEAM Olympiad 2023** – Logistics Lead
- **BUET Admission Test 2023** – Student Guide
- **Machine Learning & AI with Python** (Harvard – Online)
- **Machine Learning Mastery Course** (BUET Robotics Society)
- **Research Skill Development Workshop** (DCE, BUET)

REFERENCES

Dr. A. B. M. Mainul Bari – Associate Professor, Dept. of IPE, BUET
mainulbari@ipe.buet.ac.bd

Dr. Abdullahil Azeem – Professor, Dept. of IPE, BUET
azeem@ipe.buet.ac.bd